Chapter 4 · Solutions

P15 · Apply whole determinants

Hint. Start with the listed attributes, then fire every available rule.

(VW)+ starts {V,W}. V→X adds X; WX→Y and WX→Z add Y,Z. Final closure: VWXYZ. U is missing, so VW→U does not follow. U+ adds V,W, then X,Y,Z, and is UVWXYZ.

Without V→X, (VW)+ stays VW. X is missing, so neither WX rule can fire. U+ becomes UVW. Having W alone is not enough for a WX determinant.

P16 · Find every minimal identifier

Hint. C cannot be obtained from another attribute.

Every key contains C. C+ is CD. Either A or B supplies the missing pair A,B, giving keys AC and BC. Removing either member of either key loses full determination. Prime attributes are A,B,C.

After removing B→A, A and C are both compulsory: neither occurs on a right side. AC is the sole candidate key and A,C are prime. BC+ is BCD and misses A.

P17 · Count the union of supersets

Hint. Count each key's supersets, then subtract their overlap.

On ABCD, each two-attribute key has 2^2=4 supersets. Sets containing both keys must contain ABC, leaving D optional: 2 overlaps. Count: 4+4−2=6.

With A→E, both AC and BC still determine everything. On five attributes each has 2^3=8 supersets. Their overlap has ABC compulsory and D,E optional, giving 4. Count: 8+8−4=12.

With an unconstrained E, every key must include E. Keys become ACE and BCE. Each has four supersets and their overlap has two. Count returns to 6. More columns do not necessarily mean more optional choices.

P18 · Simplify without changing the rules

Hint. Test removals by closure after removing the rule being tested.

Split A→BC into A→B and A→C. B is extraneous in AB→D because A already obtains B, so replace it by A→D. A→C is redundant because A→B→C remains. The resulting G is {A→B,B→C,A→D}.

Each original rule follows from G, and each G rule follows from F. Every right side is a singleton. Removing any G rule loses its consequence, and no left side can be shortened. Thus G is a minimal cover.

Without A→D, A+ is ABC rather than ABCD. AB→D no longer follows, so the smaller G is not equivalent to F.

P19 · Keys before normal forms

Hint. Prime means belongs to at least one candidate key.

With AB→C,C→B, the keys are AB and AC. All attributes are prime. C+ is BC, so C is not a superkey. C→B violates BCNF, but B is prime and satisfies the 3NF exception. Highest form: 3NF.

With A→B,B→C, the only key is A. B→C has a non-superkey determinant and a nonprime dependent, so it violates 3NF. There is no partial dependency on a proper nonempty part of the one-attribute key. Highest form: 2NF.

P20 · Can the join invent rows?

Hint. For a binary split, close its common attributes.

AB/AC has common A. A+ includes AB, so it is lossless. AB/BC has common B, whose closure is B. It determines neither AB nor BC, so this split is not guaranteed lossless.

Original legal rows (a1,b1,c1),(a2,b1,c2) satisfy A→B. Their projections join into four rows; (a1,b1,c2),(a2,b1,c1) are extra. This proves the failure.

After adding B→C, B+ includes BC. The AB/BC split is now lossless. The old counterexample violates the added rule and can no longer be used.

P21 · Chase the three-way split

Hint. A present attribute gets its shared a symbol. An absent attribute gets a fresh b symbol.

Component	A	B	C	D
AB	aA	aB	b0C	b0D
BC	b1A	aB	aC	b1D
AD	aA	b2B	b2C	aD

A→B merges b2B with aB because AB and AD agree on A. B→C then makes the three C entries agree with the distinguished aC.

Component	A	B	C	D
AB	aA	aB	aC	b0D
BC	b1A	aB	aC	b1D
AD	aA	aB	aC	aD

AD is entirely distinguished, so the split is lossless. Without B→C, AD ends (aA,aB,b2C,aD). No row is entirely distinguished, and no further rule applies. The split is not guaranteed lossless.

Check. This is the FD chase, not a one-time comparison of intersections.

P22 · Preserve constraints locally

Hint. Project implied dependencies, not just the originally written ones.

AB/AC has local rules A→B and A→C. Their union gives B+={B}, so B→C is not preserved. The split is nevertheless lossless because the common A determines AB.

AB/BC has local A→B and B→C. Both original rules are enforceable without joining. Its common B determines BC, so it is also lossless. Here both properties hold.

P23 · Choose the guarantee you need

Hint. A BCNF split can be lossless without preserving every dependency.

Using C→B yields BC and AC. C is a key of BC; AC has no nontrivial projected violation. Both components are BCNF. Their common C determines BC, so the split is lossless.

Their local rules only provide C→B. Starting AB under these local rules does not obtain C, so AB→C is not preserved.

3NF synthesis produces ABC from AB→C and BC from C→B. BC is contained in ABC and can be removed. ABC contains key AB, so no extra key relation is needed. Keeping ABC is acceptable because the target is 3NF: its prime-attribute exception permits C→B. It preserves all dependencies and has no information-loss issue.